

Midterm Exam 2

Last Name _____ First Name _____ Student ID _____

Honor Code: This exam represents only my own work. I did not give or receive help from anyone.

Instructions

- You may use the instructor-provided reference sheet (double-sided). No other materials.
- Calculator permitted (non-programmable).
- Show your work. Answers without work may not receive full credit.
- Write answers in the space provided.

Part I — Multiple Choice [20 pts, 2 pts each]

Q1.	Which delay varies unpredictably with traffic load?	Queuing delay
Q2.	CIDR primarily enables:	Efficient address aggregation
Q3.	Which algorithm requires full topology knowledge?	Dijkstra
Q4.	When TTL reaches 0, the router:	Drops the packet
Q5.	UDP provides:	Best-effort delivery

Part II — Short Answer [30 pts, 6 pts each]

1. Given IP 192.168.10.34/26, find network, broadcast, and number of hosts.
2. Explain the difference between forwarding and routing.
3. Describe how NAT works.
4. Explain why Stop-and-Wait is inefficient.
5. Describe the TCP three-way handshake.

Part III — Problems [50 pts]

Q1: Subnetting (20 pts) Given 10.0.0.0/16, create 4 equal subnets and specify their CIDR blocks.

Q2: Routing (15 pts) Run Dijkstra on the given graph and compute shortest paths.

Q3: TCP (15 pts) Explain slow start and behavior after packet loss.